

Quantum Fault Tolerance Meets Toom's Contours

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Change letters: the d in the model section refers to a size that is comparable to the code distance where here d refers to the dimension of the Toric complex

1 The (d, d') -Dimensional Toric Code and the SWEEP Rule

Definition 1.1 (d -Dimensional Toric Complex). For integers d and n , denote the group resulting from the d -directed power of the cyclic group of order n by $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$, and denote the standard generating set of $\mathcal{G}_n^d$ by $S = \{1_i : i \in [d]\}$. We define the d -dimensional toric complex to be the hypergraph obtained from the Cayley graph $\text{Cay}(\mathcal{G}_n^d, S)$ by taking its vertices as the 0-faces and for any $i \in [d-1]$ we define the i -faces as follow: First, for a vertex $v \in \mathcal{G}_n^d$ and for a subset of generators $S' \subset S$ of size $|S'| = i$, the i -face rooted at v and spanned by S' , denoted by $\theta_{v,S'}$, is:

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\}$$

The i -faces of the complex are the collection of all the possible rooted spanned i -faces induced by $\text{Cay}(\mathcal{G}_n^d)$. We will denote them by $\mathcal{F}_i$.

Additionally, we name n and d , the *side length* and the *dimension* of the complex.

Definition 1.2 (Positive Edges and Cubes). For $u, v \in \mathcal{G}_n^d$, and $S'_1, S'_2 \subset [d]$, with sizes $|S'_1| = |S'_2| = i$, we say that the i -faces θ_{v,S'_1} and θ_{u,S'_2} are adjacent if the intersection $\theta_{v,S'_1} \cap \theta_{u,S'_2}$ is an $i-1$ -face. In addition, if θ_1 and θ_2 are i and $i-1$ -faces such that $\theta_2 \subset \theta_1$, then we say that θ_2 is a *positive edge relative to face* θ_1 if they are rooted at the same vertex. Furthermore, we say that θ_1 is a *positive cube* of θ_2 . When the i -face is clear from the context, we will abbreviate and say that θ_2 is *positive*.

Example 1.3. For example, let $v, u \in \mathcal{G}_n^d$ and $g_1 \neq g_2 \in S$. Consider the 2-face $\theta = \{v, g_1v, g_2g_1v, g_2v\}$. Then, for $x, y \in \mathcal{G}_n^d$, we say that an edge (1-face) $\{x, y\}$ is *positive relative to the face* θ if $x = v$ and y is either g_1v or g_2v .

We denote the complex by $G = (V, E, F)$, where V , E , and F are the 0, 1, and 2 faces.

Add a paragraph that explains that we choose not to use a chain-complex since we earn nothing from it i.e don't care about topology properties. Yet mention that chain-complex formulation proved itself as an effective tool to compute the properties of the codes we use.

Definition 1.4 (Assignment on The (d', d) -Toric Complex). Let $\mathcal{T}$ be a d -dimensional complex. Each of its i -face sets naturally induces a vector space $\mathbb{F}_2^{|\mathcal{F}_i|}$. We call a binary vector $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$ an assignment on the i -faces of $\mathcal{T}$. For a vertex v and a subset of generators $S' \subset S$ of size $|S'| = i$, we use the notation $\theta_{\mathbf{v}, S'}$ to represent the unit vector in $\mathbb{F}_2^{|\mathcal{F}_i|}$ that has 0 in every coordinate except the one associated with the i -face $\theta_{v, S'}$. Then

$$\left\{ \theta_{\mathbf{v}, S'} : v \in \mathcal{F}_0, S' \subset S \text{ s.t } |S'| = i \right\}$$

is a base to $\mathbb{F}_2^{|\mathcal{F}_i|}$ **Not exactly base, they are linear dependent.** Let z be an assignment on the i -faces. Let $v_1, \dots, v_k \in \mathcal{F}_0$ and $S_1, \dots, S_k \subset S^{\times k}$ be the generator subsets such that $\theta_{v_1, S'_1}, \theta_{v_2, S'_2}, \dots, \theta_{v_k, S'_k}$ are the collection of i -faces that support z , namely:

$$z = \sum_{i=1}^k \theta_{\mathbf{v}_i, S'_i}$$

We refer to $\theta_{v_1, S'_1}, \theta_{v_2, S'_2}, \dots, \theta_{v_k, S'_k}$ as the collection of faces that form z , or briefly, the faces form z . For $i > 0$, we define the linear map $\partial : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ such that

$$\partial \theta_{\mathbf{v}, S'} := \sum_{g \in S'} \theta_{\mathbf{v}, S' / g} + \sum_{g \in S'} \theta_{\mathbf{g} \mathbf{v}, S' / g}$$

For $i < d$, we define the linear map $\partial^* : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ such that

$$\partial^* \theta_{\mathbf{v}, S'} := \sum_{g \in S / S'} \theta_{\mathbf{v}, S' \cup g} + \sum_{g \in S'} \theta_{\mathbf{g}^{-1} \mathbf{v}, S' \cup g}$$

We define the *restriction of z to a vertex v* to be the linear map $|_v : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u}, S'}|_v := \begin{cases} \theta_{\mathbf{u}, S'} & v \in \theta_{\mathbf{u}, S'} \\ 0 & \text{else} \end{cases}$$

We define the *front of z to a vertex v* to be the linear map $|_{v+} : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u}, S'}|_{v+} := \begin{cases} \theta_{\mathbf{u}, S'} & u = v \\ 0 & \text{else} \end{cases}$$

We say that an assignment z is *connected* if the graph that is induced by taking the faces that support z as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of z to be its subsets such that each of them is a connected assignment. For a j -face f , we say that $f \in z$ if z has a support on $\theta_{\mathbf{v}, \mathbf{S}'}$ such that $f \in \theta_{\mathbf{v}, \mathbf{S}'}$. Consider an assignment $z \in \mathbb{F}_2^{\mathcal{F}_i}$ and a vertex $v \in \mathcal{G}_n^d$. We say that z is a *rooted assignment* at vertex v if there are subsets $S'_1, \dots, S'_k \subset S^{\times k}$ such that each of S'_j is at size i and z is decomposed as:

$$z = \sum_j^k \theta_{\mathbf{v}, \mathbf{S}'_j}$$

Put differently, z is the result of taking a collection of i -faces that are rooted at v .

Add a proof for $\partial\theta_{\mathbf{v}, \mathbf{S}'} \cdot \partial^*\theta_{\mathbf{u}, \mathbf{S}''} = 0 \Rightarrow$ gives a CSS code.

Need to decide on what should I elaborate, for example should I provide a proof for the distance and the rate of the code? [Den+02]

Definition 1.5 ((d, d') Toric Code). Let d, d' be integers such that $1 < d' < d$. For any integer n , we construct the n th instance of the quantum code family (d, d') *Toric Code* as follows: Let $G = (\mathcal{F}_0, \dots, \mathcal{F}_d)$ be the d -dimensional toric complex obtained from $\mathcal{G}_n^d$ as defined in Definition 1.1. We associate the (q)bits with the $\mathcal{F}_{d'}$, the X -checks with the $\mathcal{F}_{d'-1}$, and the Z -checks with the $\mathcal{F}_{d'+1}$. An X -check, c_x , is satisfied if the parity of (q)bits set on the d' -face containing c_x is even. Similarly, a Z -check, c_z , is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the d' -faces contained in c_z is even.

Definition 1.6 (Sweep Rule). Let z be an assignment over the d' -faces, The *sweep rule decoding procedure* at vertex $v \in \mathcal{G}_n^d$, defining as follow:

1. For the $\mathcal{C}_X$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial\tilde{z})|_{v+} = \partial(z)|_{v+}$. Namely, the X -checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.
2. Apply the Hadamard gate on any d' face containing v .
3. For the $\mathcal{C}_Z$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial^*\tilde{z})|_{v+} = \partial^*(z)|_{v+}$. Namely, the Z -checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.
4. Apply the Hadamard gate again on any d' face containing v to reverse the second stage.

Stages 1 and 3 involve measurement of stabilizers.

Definition 1.7 (Decoding Gadget). Consider a quantum circuit C , for a quantum gate u in C , with fun-in Δ , we say that u is a *decoding gadget* if it has the following form:

1. The wires of u wires can be decomposed into two subsets A and B such that the wires in A are reset wires, and there exists an encoded register R in C such that the wires in B are all set in R .
2. For a set of faults $\mathcal{E}$, there is a Pauli gate u' that acts only on B such that the action of u in $C[\mathcal{E}]$ equals u' . Put differently, the quantum circuit obtained from $C[\mathcal{E}]$ by replacing u with u' equals $C[\mathcal{E}]$.

Definition 1.8 (Unitary Decoding Representation). Let $C = \{u_i\}_{i=1}$ be a quantum circuit, and let $\mathcal{U} = \{u'_i\}_{i=1} \subset C$ be a subset of the gates in C that are decoding gadgets. For a set of faults $\mathcal{E}$, we define the *unitary decoding representation* of $C[\mathcal{E}]$ to be the quantum circuit obtained by replacing each of the gates in $\mathcal{U}$ with their corresponding Pauli gate according to $\mathcal{E}$.

Claim 1.9. The local gates performing Toom's rule are decoding gadgets.

Proof. [prove that.](#) □

Definition 1.10 (Registers Coordinates System). Let $C = \{u_i\}_{i=1}$ be a quantum circuit, with width w and registers $\mathcal{R} = R_1, \dots, R_k$, each of length at most n . For register R_i , we define the register identifier $\mathbf{1}_{R_i} = \mathbf{1}_i$. Then, we define the map $\phi_{\mathcal{R}}: \mathbb{Z}_w \rightarrow \mathbb{Z}_R \times \mathbb{Z}_n$ to return, for a space-location coordinate $x \in \mathbb{Z}_w$, the tuple consisting of the identifier of the register containing x , and its relative coordinate within the register. Namely,

$$\psi_{\mathcal{R}}(x) = (\mathbf{1}_{R_j}, x')$$

Where R_j is the register containing x , and x' is the index of the qubit, set at the x location, in R_j . The presentation of C in the *registers coordinate system* is the encoding by a set of gates $\{u'_j\}_{j=1}$, which is the result of replacing each space-location of the original gates with the applications of $\psi_{\mathcal{R}}$ on them. Namely, for each $u_i = (g_i, l_i) \in C$, we map its space-location coordinate $l_{i_2} = (l_{i_2}^{(1)}, \dots, l_{i_2}^{(\Delta)})$ to:

$$l_{i_2} \mapsto (\psi_{\mathcal{R}}(l_{i_2}^{(1)}), \dots, \psi_{\mathcal{R}}(l_{i_2}^{(\Delta)}))$$

For brevity, we name the first and second coordinates in the result of $\psi_{\mathcal{R}}$ the *register coordinate* and the *inner coordinate*.

[COMMENT] Choose one of the following definitions for the base graph G_o . Pitori Berman and Peter Gacs chose the second, namely the projected version, yet if I had to guess, I would say that the first gives an advantage. Furthermore, We might want to fold computation stages, such that we have GATE $\rightarrow$ NOISE $\rightarrow$ DECODING, and after each such cycle freeze the state of the graphs.

Definition 1.11 (Troic Encoded Circuit). Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and each of the registers is encoded by $\mathcal{C}$.

2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form Λ, P, D .

Definition 1.12 (Base graph G_o). Let C be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, \dots, \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \{(v, t) : t \in [d], v \in \bigcup_i \mathcal{G}_i\}$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t + 1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$E_o := \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t + 1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1\}$$

Definition 1.13 (Base graph G_o). Let $C = \{u_i\}_{i=1}^n$ be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}$ the Toric complex that induces the codes, and suppose that the locations of the gates in C are given in registers' coordinate systems. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in $\mathcal{G}$, and the second coordinate is associated with a time. Namely:

$$V_o := \{(v, t) : t \in [d], v \in \mathcal{G}\}$$

2. The edges E_o are derived from the original edges in $\mathcal{G}$, with the addition of edges of the following form: For two vertices at preceding times t and $t + 1$, let's denote them by (v_1, t) and $(v_2, t + 1)$, for which there exist registers R'_1, R'_2 , and faces $f_1, f_2 \in \mathcal{G}$, such that $v_1 \in f_1, v_2 \in f_2$ and C contains a gate that acts non-trivially on the locations $(t, f_1, \mathbf{1}_{R'_1})$ and $(t, f_2, \mathbf{1}_{R'_2})$ at time t , are also connected by an edge. Namely:

$$E_o := \{(v, t), (u, t')\} : \text{if} \\ \text{either } \{v, u\} \in \mathcal{G} \text{ and } t' = t, t + 1 \\ \text{or there exists } u_i = (g_i, l_i) \in C, f_1, f_2 \in \mathcal{G}, \text{ and } R'_1 R'_2 \\ \text{such } v \in f_1, u \in f_2, t = l_{i,1}, t' = l_{i,1} + 1 \\ \text{and } (f_1, \mathbf{1}_{R'_1}), (f_2, \mathbf{1}_{R'_2}) \in l_{i,2}\}$$

2 Infinite Torics

Definition 2.1 (Shifting in Registers System). Consider the register coordinate system induced by a quantum circuit of width w , with R registers, each of length at most n . For an integer $r \in \mathbb{Z}$, we define the r -shifting function $\text{Sh}_r: (\mathbb{Z}_R \times \mathbb{Z}_n)^{\times \Delta} \rightarrow (\mathbb{Z}_R \times \mathbb{Z})^{\times \Delta}$ to act on space-location coordinates as follows:

$$\begin{aligned} \text{Sh}_r(l_1, \dots, l_k) &= (\text{Sh}_r(l_1), \dots, \text{Sh}_r(l_k)) \\ \text{Where } \text{Sh}_r(l_i) &= \begin{cases} l_i + r & \text{if } l_i \text{ is an inner coordinate} \\ l_i & \text{else} \end{cases} \end{aligned}$$

Definition 2.2. Let n be an integer, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with registers $\mathcal{R} = R_1, \dots, R_k$ such that each register R_i is a toric encoding with a side length n . In addition, all the decoding gadgets of C are sweep rule gadgets; let us denote them by $\mathcal{U}$. We define the ∞ -extension of C , denoted by C^∞ , as follows:

1. For a register $R_i \in \mathcal{R}$, denote by d_i, d'_i the dimension of the toric complex that induces the code, and the dimension of the faces that are associated with the bits. Each register R_i in $\mathcal{R}$ is mapped to a register R'_i , which is a (d_i, d'_i) -dimensional, infinite side length, toric encoding. Since the mapping between registers is one-to-one, we use the identifier $\mathbf{1}_{R_i}$ to identify also R'_i .
2. Each gate $u_i \in C/\mathcal{U}$ is mapped to:

$$u_i = (g_i, l_i) \mapsto \bigcup_{j=-\infty}^{\infty} \{(g_i, (\text{Sh}_{jn}(l_i)))\}$$

3. Each gate $u \in \mathcal{U}$ is mapped to the collection of sweep rule gadgets $\{v_j\}_{j=-\infty}^{\infty}$ such that each of them has the same time coordinate as u , is in the register that corresponds to the image of u 's register, and is rooted at $\text{root}(v) + jn$.

For a set of faults $\mathcal{E} = \{f_i\}_{i=1}$, we denote by $C^\infty[\mathcal{E}^\infty]$ the ∞ -extension of $C[\mathcal{E}]$.

Claim 2.3. Let $u = (g, l)$ and $v_0, v_{\pm 1}$ plantation of u at locations $\text{root}(u)$ and $\text{root}(u) \pm n$. Then $v_0 v_{\pm 1}|_{[w]} = u$.

Proof. [Prove it](#) □

Definition 2.4. For a quantum circuit C , of the form defined in Definition 2.2, and a set of faults $\mathcal{E}$, denote:

$$\begin{aligned} \rho &:= C[\mathcal{E}] |0\rangle\langle 0| C[\mathcal{E}]^\dagger \\ \rho_\infty &:= C^\infty[\mathcal{E}^\infty] |0\rangle\langle 0| (C^\infty[\mathcal{E}^\infty])^\dagger \end{aligned}$$

We say that a quantum circuit C is ∞ -compatible if, for any set of faults $\mathcal{E} = \{f_i\}_{i=1}$, it holds that:

$$\rho = \mathbf{Tr}_{/[w]}(\rho_\infty)$$

Claim 2.5. Let C be quantum circuit of the form defined in Definition 2.2. Then C is ∞ -compatible.

Proof. Its enough to prove the claim for circuits in their unitary decoding representation. We prove it by induction on the number of the gates.

1. Base. For an empty circuit, ρ_∞ is just the infinite string of zeros; thus, the restriction of it to $[w]$ locations, namely $\mathbf{Tr}_{/[w]}(\rho_\infty)$, yields a single matrix element corresponds to the w -length string of zeros, which is exactly ρ .
2. Step. Assume the correction of the claim for any circuit, with less than τ steps, and consider a circuit C with τ steps. Denote by C' the circuit obtained from C by erasing the last gate in C , observes that C' is a circuit with $\tau - 1$ steps. Denote by ρ' and ρ'_∞ the: [Enter it](#). By the induction assumption we have:

$$\rho' = \mathbf{Tr}_{/[w]}(\rho'_\infty)$$

Denote the last gate in C by u , and by $I = \{v_j\}_{-\infty}^\infty$ the collection of gates in C^∞ to which u is mapped. Let us split into cases, depending on whether u is or is not a decoding gate.

- (a) Suppose that u is not a decoding gate. Let g and l be the gate element and the location of u . By the definition of ∞ -extension, I equals:

$$I = \bigcup_{j=-\infty}^{\infty} \{(g, (\text{Sh}_{jn}(l)))\}$$

So there is only one gate in I , corresponding to $j = 0$, that has support on qubits with spatial location in the range $[w]$. Denote that gate by v_0 .

$$\begin{aligned} \mathbf{Tr}_{/[w]}(\rho_\infty) &= \mathbf{Tr}_{/[w]} \left(\prod_{j=-\infty}^{\infty} v_j \rho'_\infty \prod_{j=-\infty}^{\infty} v_j^\dagger \right) \\ &= v_0 \mathbf{Tr}_{/[w]} \left(\prod_{j=-\infty, j \neq 0}^{\infty} v_j \rho'_\infty \prod_{j=-\infty, j \neq 0}^{\infty} v_j^\dagger \right) v_0^\dagger \\ &= v_0 \mathbf{Tr}_{/[w]} \left(\prod_{j=-\infty, j \neq 0}^{\infty} v_j^\dagger \prod_{j=-\infty, j \neq 0}^{\infty} v_j \rho'_\infty \right) v_0^\dagger \\ &= v_0 \mathbf{Tr}_{/[w]}(\rho'_\infty) v_0^\dagger \\ &= v_0 \rho' v_0^\dagger \\ &= \rho \end{aligned}$$

- (b) In the case that u is a decoding gate, observe that there are at most two gates in I that have support on the qubits with space-location in $[w]$. If

there is exactly one, then repeating the non-decoding gate case while setting this gate as v_0 gives the desired result. Else, in the case where two gates act non-trivially on qubits at $[w]$, denote those gates by v_0, v_1 . Since we assumed that C is given in its unitary decoding representation, it holds that v_0, v_1 are Paulis, in particular a product operator. Denote their product decompositions by $v_0 = \prod_k v_0^{(k)}$ and $v_1 = \prod_k v_1^{(k)}$. Thus, by repeating the non-decoding case, but extracting from the infinite product only the terms in v_0 and v_1 that act non-trivially at $[w]$, we get:

$$\mathbf{Tr}_{/[w]}(\rho_\infty) = v_0 v_1|_{[w]} \rho' (v_0 v_1|_{[w]})^\dagger$$

Moreover, since v_0 and v_1 are a plantation of g in locations $\text{root}(u)$ and $\text{root}(u) \pm n$, then by Claim 2.3, we have that $u = v_0 v_1|_{[w]}$; therefore:

$$\begin{aligned} \mathbf{Tr}_{/[w]}(\rho_\infty) &= v_0 v_1|_{[w]} \rho' (v_0 v_1|_{[w]})^\dagger \\ &= u \rho' u^\dagger \\ &= \rho \end{aligned}$$

□

Add that the computation graph is lifted naturally to the ∞ -extension, since the image of a mapping are overlapped gates, we can execute them simultaneously. Same for the base graph $G_o \rightarrow G_o^\infty$.